

Test Document: Solar Energy Basics

Solar energy is power obtained by capturing heat and light from the sun. This energy can be harnessed using a range of technologies, most commonly solar panels made of photovoltaic cells, which convert sunlight directly into electricity.

Solar panels work by allowing photons from sunlight to knock electrons free from atoms, generating a flow of electricity. This direct current (DC) electricity is then converted into alternating current (AC) using an inverter, making it usable for homes and businesses.

One major advantage of solar energy is that it is renewable and produces no direct emissions during operation. However, the efficiency of solar panels can be affected by weather conditions, panel angle, and geographic location.

Battery storage systems are often paired with solar panels to store excess energy generated during the day for use at night or during cloudy periods. This combination allows for greater energy independence from the traditional power grid.